

Stormflow generation in two headwater catchments in eastern Amazonia, Brazil

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Abstract:

Throughout the tropics, and the Amazon region in particular, only a few experimental studies have identified the main hydrological pathways and response to storm events. This study identifies the hydrological response patterns and quantifies the main runoff generating processes for two headwater catchments in eastern Amazonia, an area of low relief. Over an 18 month study period, 245 and 55 rainfall–runoff events at the respective headwater catchments were analysed. The rainfall–runoff regression lines for both catchments revealed a remarkably strong linear correlation between event rain total and runoff volume. The area contributing to stormflow was proven to be constant in extent at approximately 0.6% of the catchment and to coincide with the exact extent of the riparian wetland zone. The soils of the surrounding hillslopes were found to be highly permeable oxisols. Indications of secondary permeability due to the deep root system of the secondary vegetation were also found. Copyright © 2007 John Wiley & Sons, Ltd.

KEY WORDS stormflow; runoff generation; riparian forest; eastern Amazonia; Brazil

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INTRODUCTION

The generation of runoff or streamflow is the result of a series of interactions between the land surface and the water entering the system. The volume of water appearing as stormflow in the hydrograph of a stream is often a small fraction of the total runoff. The remainder of the rainwater input leaves the catchment as evapotranspiration, baseflow or deep groundwater outflow (Dingman, 1994; Ward and Robinson, 2000). The nature of the soil surface is the key factor that determines whether rainfall will infiltrate and move through the soil or result in a form of infiltration excess (Hortonian) overland flow, HOF (Horton, 1933), subsurface stormflow, SSF (Chorley, 1978) or saturation excess overland flow, SOF (Dunne and Black, 1970a, 1970b). Hewlett and Hibbert (1967), Hewlett and Nutter (1970) and Dunne (1978) suggested that under well vegetated and undisturbed soil conditions typically only a small part of a catchment area contributes directly to stormflow, and that the lower, near saturated valley slopes adjacent to the stream channel generate SOF. They also suggested that with increasing rain depth these areas tend to expand (the variable source area concept). Dunne (1978) suggested that in the presence of a good vegetation and litter cover little overland flow may be expected to occur over much of the drainage basin, except in the most extreme storms. Betson (1964) and Kirkby (1967) (in

Chorley, 1978) noted that contributing areas of overland flow occur generally at the slope base, in concavities where surface flow lines converge and in areas of thin soil cover. Beven *et al.* (1988) indicated the importance of the role of catchment geomorphology on the hydrological response.

A conceptual framework for comparing preferred stormflow pathways across the humid tropics was proposed by Elsenbeer and Vertessy (2000). They observed that the complex of soilscape and lithological properties and its interaction with the intensity of rainfall controls the hydrological response pattern of an area (Elsenbeer, 2001; Bonell, 2005). For forested and agricultural catchments throughout the tropics a wide spectrum of hydrological responses, under similar rainfall conditions, have been summarized by Bruijnzeel (1990), Bonell *et al.* (1993), Elsenbeer (2001) and Bonell (2005). At one end of the spectrum are substrates sufficiently permeable to let researchers conclude that any type of overland flow would only occur occasionally on the hillside (Bruijnzeel, 1983; Coelho Netto, 1987; Hodnett *et al.*, 1997a; Noguchi *et al.*, 1997a, 1997b; Nortcliff and Thornes, 1984; Nortcliff *et al.*, 1979; Walsh, 1980; Williams *et al.*, 1997; Williams and Melack, 1997). When overland flow does occur on such highly permeable soils, this flow tends not to contribute significantly to stormflow (Van de Giesen *et al.*, 2000; Stomph *et al.*, 2001, 2002; Masiyandima *et al.* 2003; Ajayi *et al.*, 2007). At the other extreme are soils with a sharp decrease in permeability at shallow depths where widespread saturation overland flow occurs, often as return flow from soil

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pipes (Bonell and Gilmour, 1978; Bonell *et al.*, 1981; Sinun *et al.*, 1992; Elsenbeer and Vertessy, 2000; Kinner and Stallard, 2004; Schellekens *et al.*, 2004; Biggs *et al.*, 2006).

Catchment scale field studies performed in the Amazon region suggest that the full spectrum of hillslope and catchment hydrological response patterns may be represented throughout the basin, but the lack of further studies prevents the definition of a comprehensive conceptual framework (Elsenbeer, 2001). Lateral pathways, predominantly in the form of overland flow, were reported for an ultisol-dominated catchment in western Amazonia (Elsenbeer and Lack, 1996), inferred for an ultisol–oxisol landscape in south-western Amazonia (Zimmermann *et al.*, 2006), and inferred for another ultisol–oxisol landscape in southern Amazonia (Johnson *et al.*, 2006). For several catchments with deep and permeable oxisols (ferralsol/yellow latosol) in central Amazonia, predominantly vertical pathways were inferred, feeding a more or less permanently saturated riparian zone (Nortcliff and Thornes, 1981, 1984; Lesack, 1993; Hodnett *et al.*, 1997a, 1997b). Although the construction of a conceptual framework based on soil group as the defining factor would be convenient, some studies have revealed marked variations in hydrological pathways and stormflow generation within soils of the same class (Elsenbeer, 2001; Chappell and Sherlock, 2005).

In the current study, an analysis is given of the main processes that control stormflow at two headwater catchments in eastern Amazonia. The principle objectives of this study were: (1) to evaluate the existence of a quick runoff pathway; (2) to determine at what end of the response spectrum the catchments can be placed;

(3) to reveal the relation between rainfall and stormflow volumes.

SITE DESCRIPTION

The study area encompasses two small subcatchments (10–34 ha; Figure 2) of the Cumaru catchment, located 12 km south-east of the town of Igarapé-Açu (1°11'S, 47°34'W) in the Bragança region of eastern Amazonia (Figure 1). The climate of the region is humid tropical with a mean annual precipitation of 2500 mm, and an average temperature of 26°C (type Am, according to Köppen (1936)), with a distinct dry season between October and December. Disturbances may originate from temperate systems travelling up along the east coast and squall (instability) lines form, instigated by the convergence of sea breeze and small-scale convective storms (Nimer, 1972; Griesinger and Gladwell, 1993; Molion, 1993). Small-scale convective storms cause the typical afternoon showers which form during the morning hours and precipitate around 1400 to 1600 hours local time (Nimer, 1972; Nimer, 1991; Molion, 1993). The landscape is characterized by a rolling topography (30–70 m a.m.s.l.) covered with a heterogeneous patchwork of agricultural fields and secondary forest, dissected by streams lined with riparian forest. In 2001, approximately 50% of the land cover consisted of secondary vegetation, and twelve percent of riparian forest (Wickel, 2004). The main crops grown in the region are cassava, corn, beans, passion fruit, and white pepper (Denich *et al.*, 2000). The dominant upland soil around Igarapé-Açu is a yellow latosol (Embrapa, 1999), which is a type of oxisol. Following the US Soil Taxonomy Rego *et al.* (1993) classified the soils around Igarapé-Açu as a red

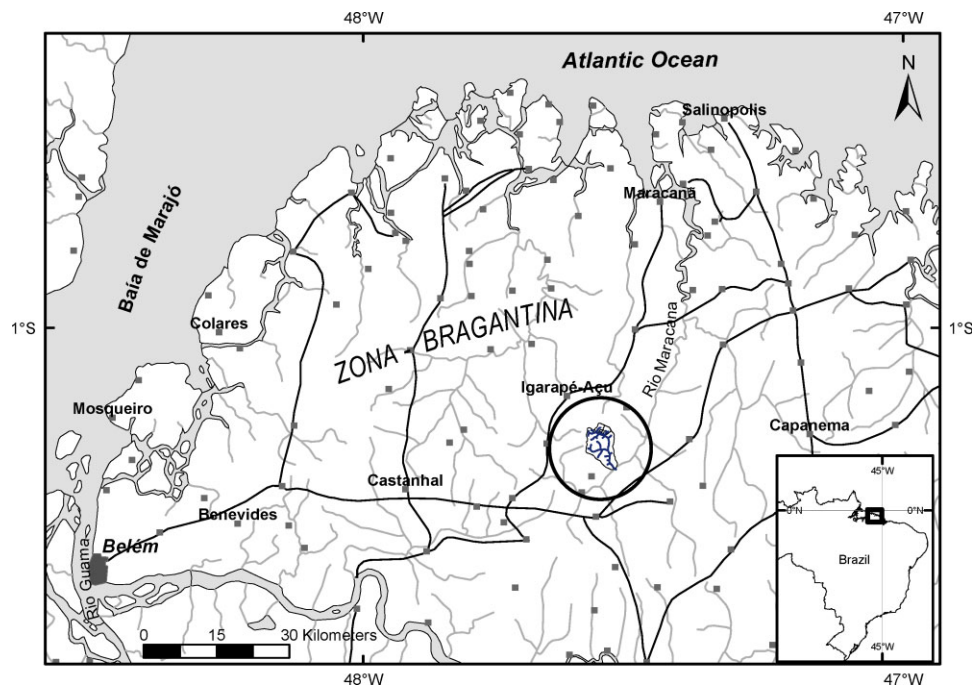


Figure 1. Location of the Cumaru watershed in the Bragança region, Brazil

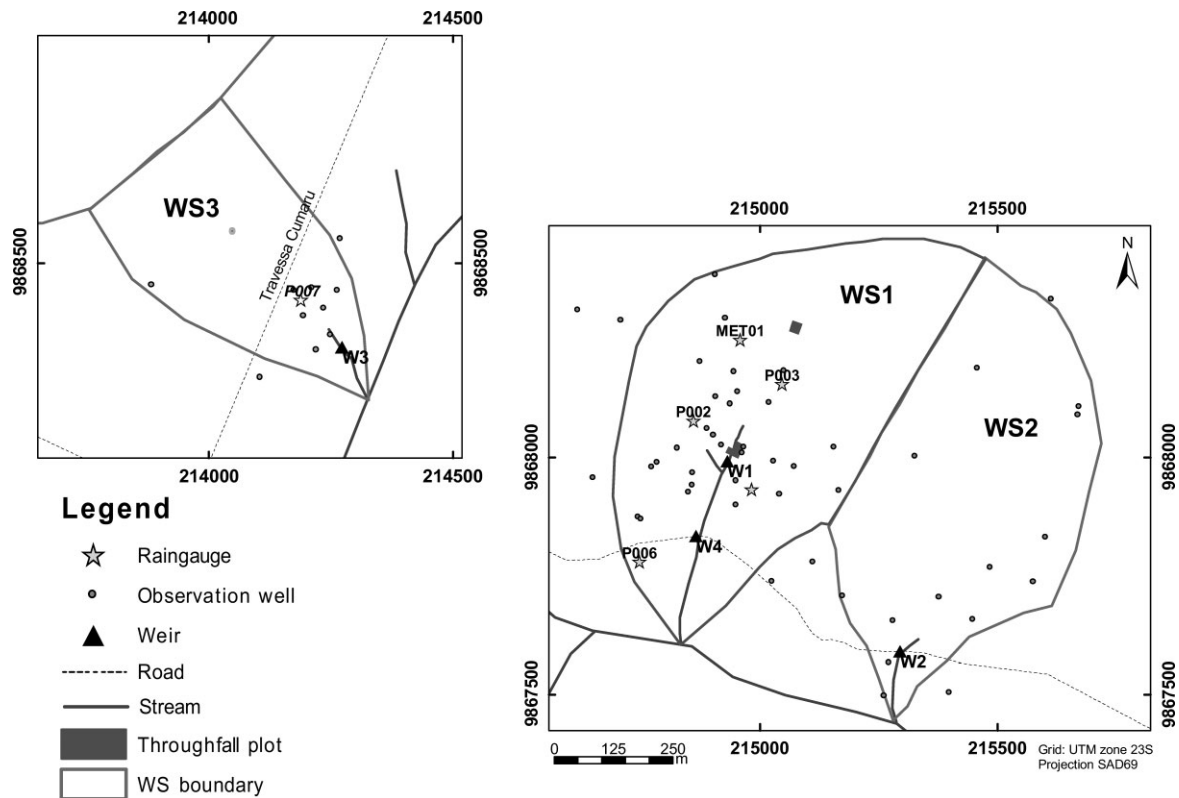


Figure 2. Location of instruments throughout the watersheds

yellow podzolic latosol (Typic Kandiudult), which is a type of ultisol. These soils are typically found in upland areas throughout the Amazon region, and were formed in the heavily weathered fluvio-lacustrine sediments of the Barreiras and post-Barreiras formations (Sombroek, 1966; Sombroek, 1984; Rossetti, 2001). These soils are moderately to highly permeable with saturated hydraulic conductivities of 150 cm d^{-1} to 250 cm d^{-1} (Sommer, 2000). The characteristics of the study area are described in more detail by Wickel (2004).

METHODOLOGY

Instrumentation

Gross precipitation (P) was measured using a tipping bucket rain gauge (Young 52 203, Traverse city, Michigan, USA; 0.1 mm per tip), and summed at 5 min intervals by a Campbell Scientific (CS) automatic weather station, Logan, Utah, USA (MET01: see Figure 2 for location), covering the study period from November 2000 to July 2002. A second tipping rain gauge (Onset RG2-M, Bourne, Michigan, USA; 0.2 mm per tip) was located at 50 m distance and generated event based totals (P003: Figure 2). An event is defined by a 5 min rainfall total greater than 0.1 mm , and is separated from a subsequent event by a minimum period of 3 h without rainfall. Three totalizing rain gauges (P002, P006 and P007: Figure 2) were distributed throughout the area in order to determine possible spatial variation in the rainfall pattern. The totalizing gauges (10 cm funnel diameter) were emptied after every major event.

The infiltrability (or infiltration capacity) of the soil was determined using a falling head double ring infiltrometer at 14 randomly stratified points throughout WS1 (cf. Kessler and Oosterbaan, 1974). First the soil inside and surrounding the ring was saturated, and the centre ring of the infiltrometer was filled with water with an initial head of 15 cm . An automatic pressure sensor logged the water level until the water level receded to the soil surface.

A blue dye tracer experiment was performed at three locations to evaluate the flowpaths of infiltrating water. A mixture of water and Brilliant Blue FCF dye was infiltrated through a metal ring with a diameter of 25 cm , which was driven 20 cm into the soil. After 20 L had infiltrated, soil material was dug out down to the wetting front as indicated by the blue dye, its depth was measured and photos of the profile were made.

Discharge measurements at WS1 were made using a 90° V-notch weir (W1) situated 100 m from the source of the stream, and a culvert 200 m downstream and close to the catchment outlet (W4) (Figure 2). The V-notch weir was equipped with an ISCO 6700C automatic water sampler which had an automatic water level sensor and a YSI multi-parameter sensor. The multi-parameter sensor measured pH, conductivity and temperature at 5 min intervals. Water level and temperature measurements at the culvert (W4) were made with a TD-DIVER pressure sensor (Van Essen Instruments, Delft, The Netherlands). Discharge measurements at WS2 and WS3 were performed with a 90° V-notch weir equipped with a TD-DIVER (W2 and W3). The TD-DIVERS logged water

level and temperature at 5 min intervals and were corrected for changes in air pressure with a barometric pressure sensor logging at the same interval.

In the current analysis streamflow data for catchments 1 and 3 (Figure 2; WS1 and WS3) are presented. WS1 was the catchment with the longest and most complete data record, running from 15 January 2001 until 16 June 2002. For WS3 observations were made between 1 February 2001 and November 2001. For W2 and W4 insufficient data were recorded for a meaningful analysis due to theft of equipment.

Hydrograph separation

Various methods exist to identify what fraction of the hydrograph consists of stormflow or baseflow. Traditionally these methods separate baseflow from stormflow by drawing a horizontal line, a line of constant slope to the inflection point (Figure 3), or a composite line (Chapman, 1999; Ward and Robinson, 2000).

The starting point of the runoff event is given by the time when the discharge (Q) starts to increase. The endpoint is commonly taken as the time when a plot of $\log Q$ against time becomes a straight line on the receding limb of the hydrograph. Using the method of Hewlett and Hibbert (1967) stream discharge can be separated into two components:

$$Q(t) = Q_{b(t)} + Q_{s(t)} \quad (1)$$

in which Q is the total discharge, Q_b represents the baseflow fraction and Q_s stormflow.

During dry periods Q_b is equal to total Q , but during storm events baseflow is assumed to increase at the rate

$$r = \frac{Q_b}{dt} \quad (2)$$

in which the slope r is considered to be a characteristic of the particular study site.

Hewlett and Hibbert (1967) suggested a slope of $0.0472 \text{ mm h}^{-1} \text{ d}^{-1}$ for small forested headwater catchments in humid regions underlain by deep soils with a high storage capacity. From the recession curves for the catchments of the current study, a much smaller slope of $0.0044 \text{ mm h}^{-1} \text{ d}^{-1}$ was found, which was close to the

value used by Lesack (1993) for a catchment with similar characteristics in central Amazonia. Typical stormflow events lasted for only 1–2 h, which makes the analysis relatively insensitive to the exact value of changes in baseflow during an event. The resulting baseflow component as a function of time during the course of a storm can then be calculated as

$$Q_{b(t)} = Q(t_i) + r(t - t_i) \quad (3)$$

where $Q(t_i)$ is the flow at the start of the storm. The end of the storm, t_f , is defined as the time when $Q(t)$ is equal to $Q_{b(t)}$. The total volume of water contributed by baseflow in a given storm is

$$V_b = \int_{t_i}^{t_f} Q_{b(t)} dt \quad (4)$$

and the volume contributed by stormflow is

$$V_s = \int_{t_i}^{t_f} Q(t) dt - V_b \quad (5)$$

RESULTS AND DISCUSSION

Rainfall

During the period 3 January to 31 December 2001, 263 separate rain events were recorded. Of these, 73% were less than 10 mm (Figure 4a), and 91% were of duration shorter than 3 h. No storms lasting longer than 11 h were recorded. Interestingly, the 32 events measured over this period that yielded more than 20 mm of rainfall accounted for 53% of the total annual amount (Figure 4a). The largest rainfall event of 2001 yielded 84.8 mm in 565 min on 11 February. About 52% of all rainfall occurred between 12:00 and 18:00 representing 60% of total annual rainfall (Figure 4b). The highest hourly rainfall intensities during the experiment amounted to 43.3 mm h^{-1} for 2001 on 17 September, and 52.5 mm h^{-1} for the first half of 2002 on 4 May. The average rate of rainfall over 2001 was 6.7 mm h^{-1} , with an average storm size of 7.7 mm, and an average duration of only 69 min.

The 2253 mm of rainfall measured over the year 2001 was close to the long-term regional average of 2400 mm (Embrapa, 1977). Over the first half of 2002, 1854 mm of rain was measured, versus 1784 mm for the same period in 2001. This indicates that the first half of 2002 did not deviate from the normal pattern. Spatial variation over the studied catchments was shown to be minimal (Wickel, 2004). The patterns observed in the current study match the climatological description given in various reviews of the Amazonian climate (Nimer, 1972; Nimer, 1991; Molion, 1993).

Infiltration

Figure 5 compares the measured K_{sat} values derived from 14 double ring infiltrometer measurements distributed over WS1 with the maximum measured rainfall rate over the study period (January 2001–June 2002).

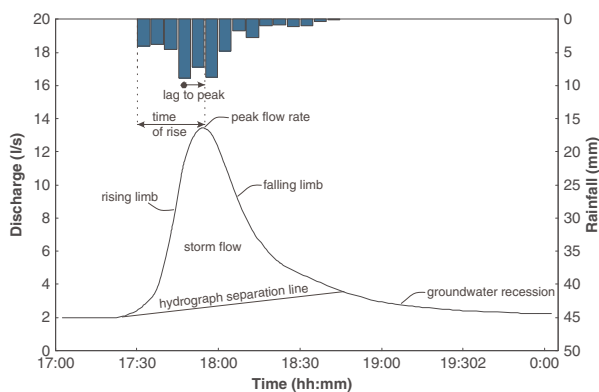


Figure 3. Components of a storm hydrograph (adapted from Hewlett, 1982)

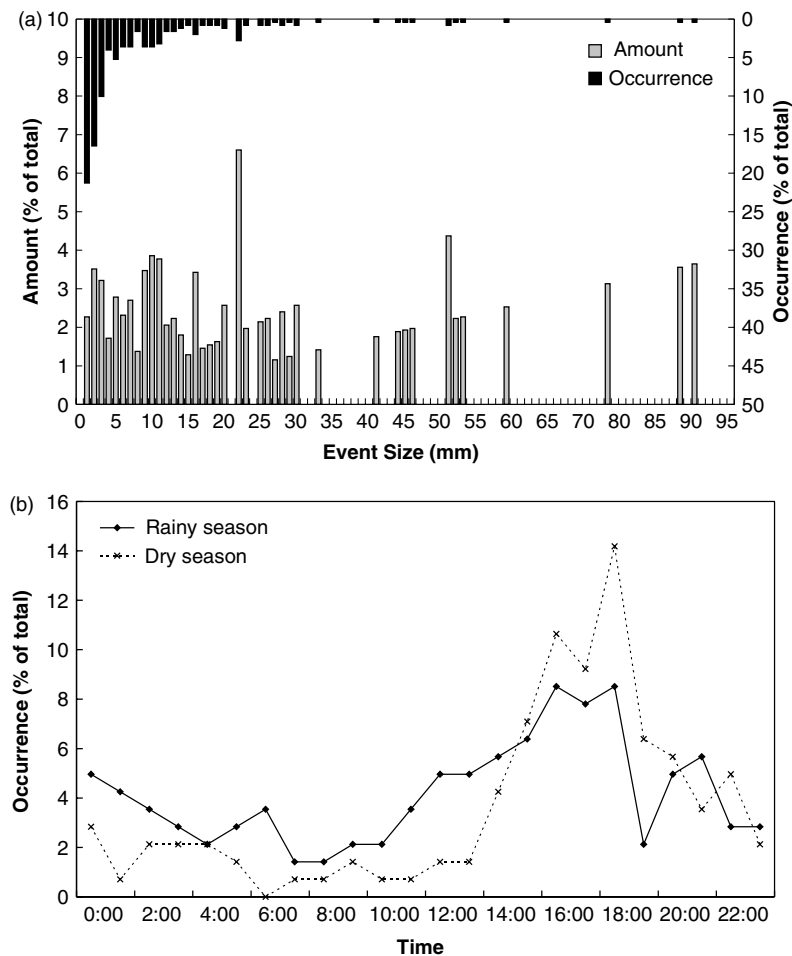


Figure 4. (a) Distribution of occurrence and relative contribution to total annual rainfall by storm events; and (b) diurnal distribution of occurrence and relative amounts of rainfall for the rainy and the dry season

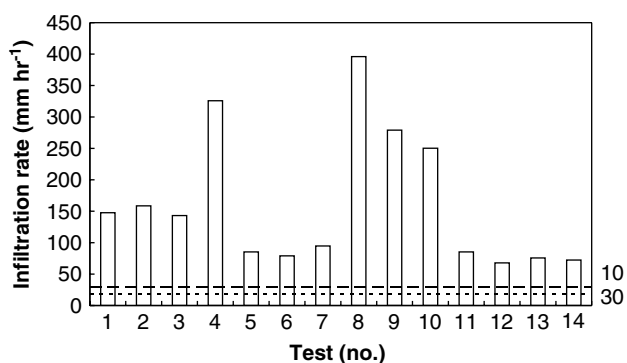


Figure 5. Infiltration rates as determined by double ring infiltrometer measurements at 14 points in WS1 (in mm h^{-1}). The dashed reference lines indicate the rainfall intensities that have a recurrence interval of 10 and 30 times per year respectively

The maximum rainfall rate observed over the study period of 52.3 mm h^{-1} never exceeded the final infiltration rates at any of the points. The average final infiltration rate obtained from the 14 measurements was 161 mm h^{-1} (386 cm d^{-1}). This value compares very well with reported values of K_{sat} of yellow latosols in the Barreiras formation in central Amazonia by Medina and Leite (1985) (545 cm d^{-1} under secondary forest), and Nortcliff and Thornes (1989) ($156\text{--}322 \text{ cm d}^{-1}$).

Using a ring permeameter (area 314 cm^2) down to a depth of 1.35 m Tomasella and Hodnett (1996) estimated K_{sat} of the top 1 m of the profile to be greater than 408 cm d^{-1} and reported extreme values as high as 2328 cm d^{-1} . They attributed this high conductivity to the presence of macropores. Below this depth, they found K_{sat} to become more dominated by particle size distribution and macro- and mesopore effects were negligible (Tomasella and Hodnett, 1996).

Sommer *et al.* (2003) estimated K_{sat} for the top 5 cm of an almost identical soil at a site approximately 2 km from WS1 with the Rosetta Programme (US Salinity Lab., Riverside, CA) and optimized their estimates by an inverse model solution with HYDRUS-1D (U.S. Salinity Lab., Riverside, CA). This model simulation supports the suggestion of a relatively homogenous soil profile down to 10 m depth with high hydraulic conductivities. The model resulted in an estimate for K_{sat} of 254 cm d^{-1} for the top 5 cm of the profile, and 160 cm d^{-1} to a depth of 10 m.

The dye experiments confirmed the high infiltration rates and revealed that preferential flow occurs along the existing root system of the fallow vegetation. A volume of 20 L of water, equalling approximately 100 mm precipitation, infiltrated through the ring in all three tests

within approximately 15 min. The infiltration front was exhumed immediately after the application of the water was finished, and found to have reached a depth between 1 and 1.5 m, partially following old root channels. Previous studies on plots close to the study area by Hölscher (1995) and Sommer *et al.* (2003) suggested deep soil-water uptake under secondary and primary vegetation in eastern Amazonia. The deep root networks described in these studies and others throughout the central and eastern Amazon region (Nepstad *et al.*, 1994; Jipp *et al.*, 1998) could provide a possible secondary conductivity, enhancing the permeability of the top soil.

Streamflow and groundwater

The hourly rainfall–runoff record at W1 is shown in Figure 6. The slow rise and fall of the baseflow over the year, and the spiked response of stormflow are evident. Comparison with the groundwater levels of three wells along a transect perpendicular to the stream revealed that the baseflow is directly correlated to the groundwater level. The strongest correlation ($r^2 = 0.88$)

between groundwater level and baseflow volume was found for the observation wells closest to the stream, and decreased gradually with distance.

Stormflow

Figure 7a shows typical responses in stream discharge at weirs W1 and W3 resulting from a complex storm

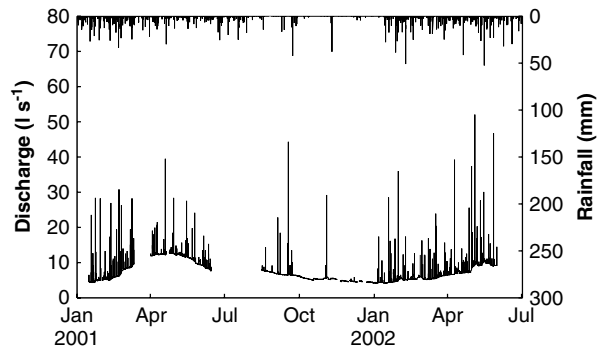


Figure 6. Rainfall and runoff record at W1 for the study period between January 2001 and June 2002

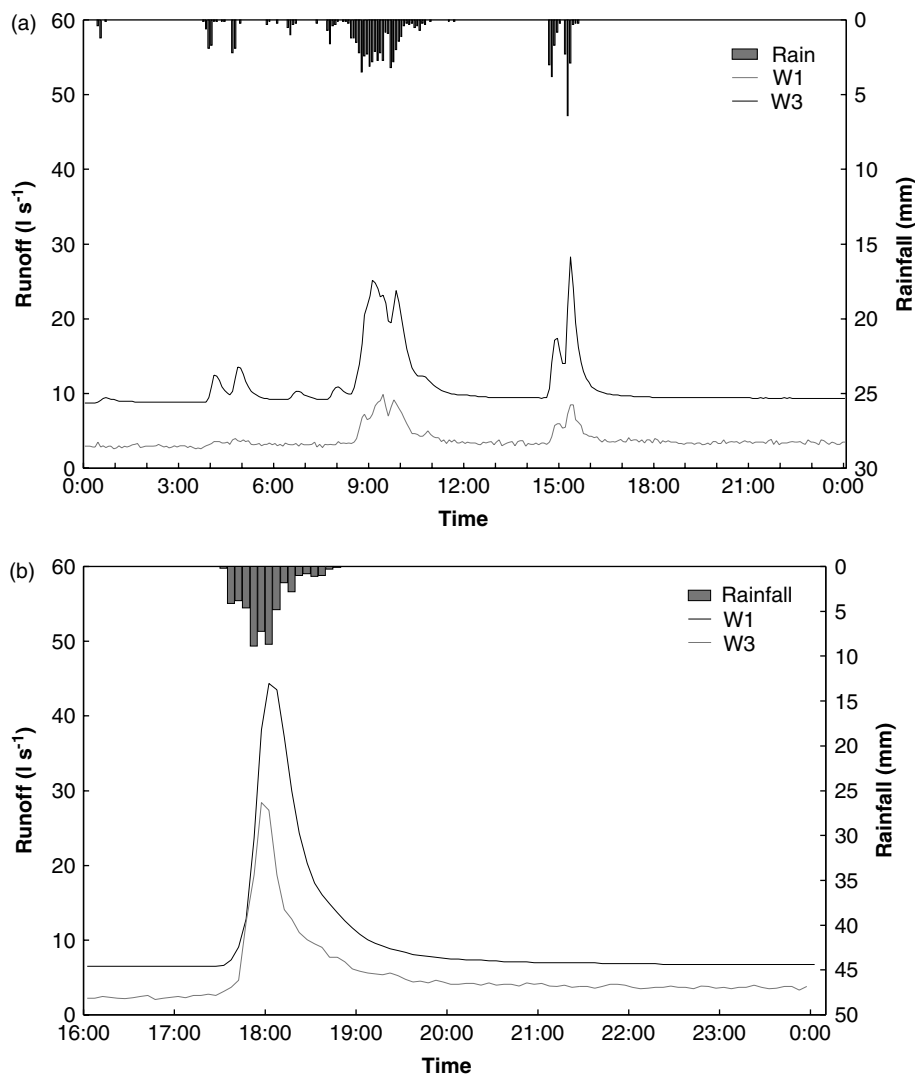


Figure 7. Runoff hydrograph at W1 and W3 resulting from two rainfall events of (a) 84 mm rainfall on 9 March 2001, and (b) 51 mm on 17 September 2001

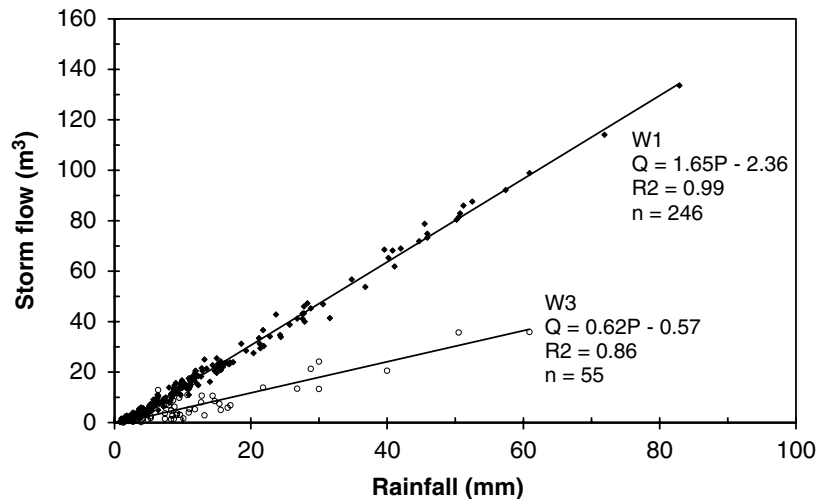


Figure 8. Rainfall total versus storm runoff for 245 storm events at W1, and 55 storm events at W3

event with an event rainfall total of 84 mm on 9 March 2001 and Figure 7b from a single storm event on 17 September 2001 of 51 mm total. As illustrated by these figures the stream responds quickly and reaches peakflow within 5 to 10 min after the highest rainfall intensity (lag-to-peak; Figure 3). Recession to the preceding baseflow level is rapid (<3 h). Given the fact that baseflow changes so slowly in comparison with the peakflows, the hydrograph separation method is efficient and is not sensitive to the exact choice of baseflow recession rate.

Over the 18 month study period 245 storm events were analysed for W1 and 55 storm events for W3. For each event yielding more than 1 mm of rain, baseflow was separated from stormflow with the linear separation technique. Both catchments (Figure 8) reveal a remarkably strong linear correlation between event rain total and runoff volume, suggesting a single and non-expanding source area for storm runoff.

The first storms of the rainy season plotted below the regression line, suggesting this is caused by the filling of the upstream valley bottom reservoir after a 3 month dry season. After this storage is filled it remains constant until it is gradually depleted again during the subsequent dry season.

The slope of the linear regression provides an estimate of the maximum area contributing to the generation of stormflow (Dickinson and Whiteley, 1970). In the case of W1, this is approximately 1650 m², and for W3 about 620 m². The surface area of the upstream section of the valley wetland as calculated from the digital elevation model (DEM), by delineating the valley bottom, is approximately 1670 m² for WS1 and 690 m² for WS3.

From this analysis, it must be concluded that stormflow is generated entirely by this area, and that the stormflow is generated in the form of SOF from the riparian wetland area. Even in the largest observed events, the source area remained the same in size. The total catchment areas contributing to W1 and W3 are 25.5 ha, and 11 ha respectively. This means that only 0.6% of both catchments functions as a stormflow source area. Everywhere

in the remainder of the catchment, rainwater infiltrates. The high infiltration rates and field observations during large storm events support this hypothesis.

The results of several catchments with comparable topographical and soil characteristics in central Amazonia (Nortcliff and Thornes, 1984; Coelho Netto, 1987; Lesack, 1993; Williams *et al.*, 1997) and Guyana (Jetten, 1994) indicate similar stormflow. Lesack (1993), Jetten (1994) and Williams *et al.* (1997) concluded that the only part of their respective catchments generating stormflow was the saturated wetland surrounding the creek and despite high rainfall intensities rainfall infiltrated, and that baseflow was the dominant pathway for runoff. Lesack (1993) estimated that up to 4% of his 23.4 ha forested catchment participated in the generation of stormflow. In none of these studies was the relation between rainfall and stormflow so strong, that even the filling of the wetland storage at the onset of the rainy season could be observed, as in the present case. It seems that the presently observed relation originates in the topography of the study catchment which consists of gently undulating terrain with sharply incised canyons, and a drainage pattern that, in turn, is the result of the hydraulic characteristics of the soils.

CONCLUSIONS

The analysis of 245 and 55 rainfall–runoff events over the 1 year study period at two headwater catchments in eastern Amazonia revealed a strong linear correlation between rainfall and stormflow volume. The correlation was found to be so strong that the fraction of the catchment accounting for stormflow runoff could be calculated. For both catchments this area was found to coincide with the extent of the riparian wetland and never exceeded 0.6%, even for the largest storms. Clearly, the storm response of these catchments is generated by direct runoff from the saturated source areas. These source areas do not vary over time and, in this, the catchments show a

storm response that differs from the variable source area runoff found in catchments in moderate climates.

The suggestion that all rainfall outside the riparian valley infiltrates and does not reach the stream channel to contribute to stormflow was supported by infiltration measurements and the strong correlation between groundwater levels and baseflow. It seems that the observed response of these catchments is mainly forced by the high permeability of the soils and possibly by secondary permeability along the deep root system of the secondary vegetation. The combination of the prevailing rainfall regime and soil characteristics has produced the gently undulating geomorphology of the region, with valleys several metres deep, incised where the groundwater interfaces with the surface. One of the objectives of this study was to determine to what extent hillslope near-surface overland flow contributed to stormflow generation. For these watersheds in the Bragança region of eastern Amazonia, this contribution is zero or negligible.

These riparian wetlands often remain the last to be forested along these valleys in the agriculturally developed areas of the Amazon region, and play an important role in the generation of stormflow. Further investigation of the stormflow-generating mechanisms throughout the Amazon region and the role of riparian forests in securing dry season flows is needed. Because of their important role in the generation of stormflow they deserve to remain conserved in order to minimize erosion and sedimentation problems.

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REFERENCES

Ajayi AE, Van de Giesen N, Vlek P. 2007. A numerical model for simulating Hortonian overland flow on tropical hillslopes with vegetation elements. *Hydrological Processes*, DOI: 10.1002/hyp.6665.

- Betson RP. 1964. What is watershed runoff? *Journal for Geophysical Research* **69**: 1541–1552.
- Beven KJ, Wood EF, Sivapalan M. 1988. On hydrological heterogeneity—catchment morphology and catchment response. *Journal of Hydrology* **100**: 353–357.
- Biggs TW, Dunne T, Muraoka T. 2006. Transport of water, solutes and nutrients from a pasture hillslope, southwestern Brazilian Amazon. *Hydrological Processes* **20**: 2527–2547.
- Bonell M. 2005. Runoff generation in tropical forests. In *Forests, Water and People in the Humid Tropics*, Bonell M, Bruijnzeel LA (eds). Cambridge University Press: 314–406.
- Bonell M, Gilmour DA. 1978. The development of overland flow in a tropical rainforest catchment. *Journal of Hydrology* **39**: 365–382.
- Bonell M, Gilmour DA, Sinclair DF. 1981. Soil hydraulic properties and their effect on surface and subsurface water transfer in a tropical rainforest catchment. *Hydrological Sciences Bulletin* **26**: 1–18.
- Bonell M, Hufschmidt MM, Gladwell JS. 1993. *Hydrology and water management in the humid tropics. Hydrological research issues and strategies for water management*. Cambridge University Press/UNESCO: Cambridge.
- Bruijnzeel LA. 1983. Evaluation of runoff sources in a forested basin in a wet monsoonal environment: a combined hydrological and hydrochemical approach. *IAHS Publications* **140**: 165–174.
- Bruijnzeel LA. 1990. *Hydrology of moist tropical forests and effects of conversion: A state of knowledge review*. Vrije Universiteit: Amsterdam, The Netherlands: UNESCO Paris.
- Chapman TG. 1999. A comparison of algorithms for stream flow recession and baseflow separation. *Hydrological Processes* **13**: 701–714.
- Chappell NA, Sherlock MD. 2005. Contrasting flow pathways within tropical forest slopes of Ultisol soils. *Earth Surface Processes and Landforms* **30**: 735–753.
- Chorley RJ. 1978. The hillslope hydrological cycle. In *Hillslope Hydrology*, Kirkby M (ed). John Wiley and Sons: Chichester; 1–42.
- Coelho Netto AL. 1987. Overland flow production in a tropical rainforest catchment: the role of litter cover. *CATENA* **14**: 213–231.
- Denich M, Kanashiro M, Vlek PLG. 2000. The potential and dynamics of carbon sequestration in traditional and modified fallow systems of the Eastern Amazon region, Brazil. In *Global climate change and tropical ecosystems*, Lal R, Kimble JM, Stewart BA (eds). CRC Press: Boca Raton, FL; 213–229.
- Dickinson WT, Whiteley H. 1970. Watershed areas contributing to runoff. *IAHS Publications* **96**: 12–26.
- Dingman SL. 1994. *Physical Hydrology*. Pearson Education Ltd.: London.
- Dunne T. 1978. *Field Studies of Hillslope Flow Processes*. John Wiley and Sons: Chichester.
- Dunne T, Black RD. 1970a. An experimental investigation of runoff production in permeable soils. *Water Resources Research* **6**: 179–191.
- Dunne T, Black RD. 1970b. Partial area contributions to storm runoff in a small New England watershed. *Water Resources Research* **6**: 1269–1311.
- Elsenbeer H. 2001. Hydrologic flowpaths in tropical rainforest soils: a review. *Hydrological Processes* **15**: 1751–1759.
- Elsenbeer H, Lack A. 1996. Hydrometric and hydrochemical evidence for fast flowpaths at La Cuenca, Western Amazonia. *Journal of Hydrology* **180**: 237–250.
- Elsenbeer H, Vertessy RA. 2000. Stormflow generation and flowpath characteristics in an Amazonian rainforest catchment. *Hydrological Processes* **14**: 2367–2381.
- Embrapa. 1977. Boletim Agrometeorológico Ano 1977—Ano 2002. EMBRAPA—CPATU: Belém, Brazil.
- Embrapa. 1999. Sistema Brasileiro de classificação de solos. EMBRAPA: Brasília, Brazil.
- Griesinger B, Gladwell JS. 1993. Hydrology and water resources of tropical Latin America and the Caribbean. In *Hydrology and Water Management in the Humid Tropics. Hydrological Research Issues and Strategies for Water Management*, Bonell M, Hufschmidt MM, Gladwell JS (eds). Cambridge University Press/UNESCO: Cambridge; 84–98.
- Hewlett JD, Hibbert AR. 1967. Factors affecting the response of small watersheds to precipitation in humid regions. In *Forest Hydrology*, Sopper WE, Lull HW (eds). Pergamon Press: Oxford; 275–290.
- Hewlett JD, Nutter WL. 1970. The varying source area of streamflow from upland basins. In *Symposium of Interdisciplinary Aspects of Watershed Management*. American Society of Civil Engineers: Bozeman, MT.
- Hodnett MG, Vendrame I, Filho ADOM, Oyama MD, Tomasella J. 1997a. Soil water storage and groundwater behaviour in a catenary

- sequence beneath forest in central Amazonia. II. Floodplain water table behaviour and implications for streamflow generation. *Hydrology and Earth System Sciences* **1**: 279–290.
- Hodnett MG, Vendrame I, Filho ADOM, Oyama MD, Tomasella J. 1997b. Soil water storage and groundwater behaviour in a catenary sequence beneath forest in central Amazonia: I. Comparisons between plateau, slope and valley floor. *Hydrology and Earth System Sciences* **1**: 265–277.
- Hölscher D. 1995. *Wasser- und Stoffhaushalt eines Agrarökosystems mit Waldbrache im östlichen Amazonasgebiet*. PhD thesis, Georg-August-Universität, Göttingen, Germany.
- Horton RE. 1933. The role of infiltration in the hydrologic cycle. *Transactions of the American Geophysical Union* **14**: 446–460.
- Jetten VG. 1994. *Modelling the effects of logging on the water balance of a tropical rainforest*. PhD thesis, University of Utrecht, Utrecht, The Netherlands.
- Jipp PH, Nepstad DC, Cassel DK, Reis De Carvalho C. 1998. Deep soil moisture storage and transpiration in forests and pastures of seasonally-dry Amazonia. *Climate Change* **39**: 395–412.
- Johnson MS, Lehmann J, Guimarães Couto E, Novães Filho JP, Riha SJ. 2006. DOC and DIC in flowpaths of Amazonian headwater catchments with hydrologically contrasting soils. *Biogeochemistry* **81**: 45–57.
- Kessler J, Oosterbaan R. 1974. Determining hydraulic conductivity of soils. *ILRI Drainage Principles and Applications* **16**: 253–296.
- Kinner DA, Stallard RF. 2004. Identifying storm flow pathways in a rainforest catchment using hydrological and geochemical modelling. *Hydrological Processes* **18**: 2851–2875.
- Kirkby M, Chorley RJ. 1967. Throughflow, overland flow and erosion. *Bulletin of the International Association of Scientific Hydrology* **12**: 5–21.
- Köppen W. 1936. Das Geographische System der Klimate. In *Handbuch der Klimatologie*, W. K. Geiger I (eds).
- Lesack LFW. 1993. Export of nutrients and major ionic solutes from a rain forest catchment in the central Amazon basin. *Water Resources Research* **29**: 743–758.
- Masiyandima MC, Van de Giesen N, Diatta S, Windmeijer PN, Steenhuis TS. 2003. The hydrology of inland valleys in the sub-humid zone of west africa: runoff processes in the M'bé experimental watershed. *Hydrological Processes*, **17**: 1213–1225.
- Medina BF, Leite JA. 1985. Influencia de tres sistemas de manejo e duas coberturas vegetais na infiltração de água em um latossolo amarelo em Manaus-AM. *Pesquisa Agropecuária Brasileira* **20**: 1323–1331.
- Molion LCB. 1993. Amazonian rainfall and its variability. In *Hydrology and Water Management in the Humid Tropics. Hydrological Research Issues and Strategies for Water Management*, Bonell M, Hufschmidt MM, Gladwell JS (eds). Cambridge University Press/UNESCO: Cambridge; 99–111.
- Nepstad DC, de Carvalho CR, Davidson EA, Jipp PH, Lefebvre PA, Negreiros GH, da Silva ED, Stone TA, Trumbore SE. 1994. The role of deep roots in the hydrological and carbon cycles of Amazonian forests and pastures. *Nature* **372**.
- Nimer E. 1972. Climatologia da Região Norte. *Revista Brasileira de Geografia* **34**: 124–153.
- Nimer E. 1991. Clima. In *Geografia do Brasil. Região Norte*, IBGE.
- Noguchi S, Rahim Nik A, Kasran B, Tani M, Sammori T, Morisada K. 1997a. Soil physical properties and preferential flow pathways in tropical rain forest, Bukit Tarek, Peninsular Malaysia. *Journal of Forest Research* **2**: 115–120.
- Noguchi S, Rahim Nik A, Yusop Z, Tani M, Sammori T. 1997b. Rainfall-runoff responses and roles of soil moisture variations to the response in tropical rain forest, Bukit Tarek, Peninsular Malaysia. *Journal of Forest Research* **2**: 125–132.
- Nortcliff S, Thornes JB. 1981. Seasonal variations in the hydrology of a small forested catchment near Manaus, Amazonas, and the implications for its management. In *Tropical Agricultural Hydrology*, Lal R, Russell EW (eds). John Wiley and Sons; 37–57.
- Nortcliff S, Thornes JB. 1984. Floodplain response of a small tropical stream. In *Catchment Experiments in Fluvial Hydrology*, Burt TP, Walling DE (eds). Geo-books: Norwich, UK; 73–85.
- Nortcliff S, Thornes JB. 1989. Variations in soil nutrients in relation to soil moisture status in a tropical forested ecosystem. In *Mineral Nutrients in Tropical Forest and Savanna Ecosystems*, Proctor J (ed). Blackwell: Oxford; 43–54.
- Nortcliff S, Thornes JB, Waylen MJ. 1979. Tropical forest systems: a hydrological approach. *Amazoniana* **6**: 557–568.
- Rego RS, da Silva BNR, Junior RSO. 1993. Detailed soil survey in an area in the municipality Igarapé-Açu. In *Summaries and Lectures Presented at the 1st SHIFT Workshop*. EMBRAPA-CPATU: Belém, PA, Brazil.
- Rossetti DF. 2001. Estratigrafia da sucessão sedimentar Posbarreiras (Zona Bragantina, Para) com base em radar de penetração no solo. *Revista Brasileira de Geofísica* **19**: 113–130.
- Schellekens J, Scatena FN, Bruijnzeel LA, van Dijk AIJM, Groen MMA, van Hogeand RJP. 2004. Stormflow generation in a small rainforest catchment in the Luquillo Experimental Forest, Puerto Rico. *Hydrological Processes* **18**: 505–530.
- Sinun W, Wong WM, Douglas I, Spencer T. 1992. Throughfall, Stemflow, Overland Flow and Throughflow in the Ulu Segama Rain Forest, Sabah, Malaysia. *Philosophical Transactions: Biological Sciences* **335**: 389–395.
- Sombroek WG. 1966. *Amazon soils. A reconnaissance of the soils of the Brazilian Amazon region*. PhD thesis, Landbouw Universiteit Wageningen.
- Sombroek WG. 1984. Soils of the Amazon region. In *The Amazon: Limnology and Landscape Ecology of a Mighty Tropical River and its Basin*, Sioli H (ed). W. Junk Publishers: Dordrecht, The Netherlands; 521–535.
- Sommer R. 2000. *Water and nutrient balance in deep soils under shifting cultivation with and without burning in the Eastern Amazon*. PhD thesis, Georg-August-Universität, Göttingen, Germany.
- Sommer RHF, Vielhauer K, Maklout Carvalho EJ, Vlek PLG. 2003. Deep Soil Water Dynamics and Depletion by Secondary Vegetation in the eastern Amazon. *Soil Science Society of America Journal* **67**: 1672–1686.
- Stomph TJ, De Ridder N, Steenhuis TS, Van de Giesen NC. 2002. Scale effects of Hortonian overland flow and rainfall-runoff dynamics: laboratory validation of a process based model. *Earth Surface Processes and Landforms*, **27**: 847–855.
- Stomph TJ, De Ridder N, Van de Giesen N. 2001. A flume design for the study of slope length effects on runoff. *Earth Surface Processes and Landforms*, **26**: 647–655.
- Tomasella J, Hodnett MG. 1996. Soil hydraulic properties and van Genuchten parameters for an oxisol under pasture in central Amazonia. In *Amazonian Deforestation and Climate*, Gash JHC, Nobre CA, Roberts JM, Victoria RL (eds). Wiley: Chichester; 101–124.
- Van de Giesen N, Stomph TJ, De Ridder N. 2000. Scale effects on Hortonian overland flow and rainfall-runoff dynamics in a West African catena landscape. *Hydrological Processes*, **14**: 165–175.
- Walsh RPD. 1980. Runoff processes and models in the humid tropics. *Zeitschrift für Geomorphologie, Supplement* **36**: 176–202.
- Ward RC, Robinson M. 2000. *Principles of Hydrology*. McGraw-Hill: Maidenhead, UK.
- Wickel AJ. 2004. *Water and nutrient dynamics of a humid tropical agricultural watershed in Eastern Amazonia*. PhD thesis, Friedrich-Wilhelms-Universität, Bonn, Germany; 122.
- Williams MR, Fisher TR, Melack JM. 1997. Solute dynamics in soil water and groundwater in a central Amazon catchment undergoing deforestation. *Biogeochemistry* **38**: 303–335.
- Williams MR, Melack JM. 1997. Solute export from forested and a partially deforested catchments in the central Amazon. *Biogeochemistry* **38**: 67–102.
- Zimmermann B, Elsenbeer H, De Moraes JM. 2006. The influence of land-use changes on soil hydraulic properties: Implications for runoff generation. *Forest Ecology and Management* **222**: 29–38.